

Python Programming

Unit 02 – Lecture 01: Lists Fundamentals and Comprehensions

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Repository: <https://github.com/tali7c/Python-Programming>

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- Create lists and access elements using indexing and slicing

Learning Outcomes

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- Apply core list methods for insertion, deletion, and sorting

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- Create lists and access elements using indexing and slicing
- Apply core list methods for insertion, deletion, and sorting
- Write list comprehensions for concise transformations
- Explain and use nested list structures

What is a List?

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- Mutable (can change after creation)
- Allows duplicates and mixed types
- Ideal for sequences you need to update

Creating Lists

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- Constructor: `letters = list("python")`
- From range: `nums = list(range(5))`
- Mixed types: `items = ["A", 3.5, True]`

Indexing

- Zero-based indexing: first element at index 0

```
scores = [80, 92, 75, 88]
print(scores[0])    # 80
print(scores[-1])   # 88
```


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- Negative indexing counts from the end

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Indexing

- Zero-based indexing: first element at index 0
- Negative indexing counts from the end
- Example: `scores = [80, 92, 75, 88]`

```
scores = [80, 92, 75, 88]
print(scores[0])    # 80
print(scores[-1])   # 88
```

Slicing

■ Syntax: `list[start:stop:step]`

```
nums = [1, 2, 3, 4, 5, 6]
print(nums[1:4])      # [2, 3, 4]
print(nums[::2])      # [1, 3, 5]
print(nums[::-1])     # [6, 5, 4, 3, 2, 1]
```

Slicing

- Syntax: `list[start:stop:step]`
- Stop index is excluded

```
nums = [1, 2, 3, 4, 5, 6]
print(nums[1:4])      # [2, 3, 4]
print(nums[::2])      # [1, 3, 5]
print(nums[::-1])     # [6, 5, 4, 3, 2, 1]
```

Slicing

- Syntax: `list[start:stop:step]`
- Stop index is excluded
- Useful for sublists and stepping

```
nums = [1, 2, 3, 4, 5, 6]
print(nums[1:4])      # [2, 3, 4]
print(nums[::2])      # [1, 3, 5]
print(nums[::-1])    # [6, 5, 4, 3, 2, 1]
```

Core List Methods

Method	Purpose
<code>append(x)</code>	Add to end
<code>extend(iter)</code>	Add all items
<code>insert(i, x)</code>	Insert at index
<code>remove(x)</code>	Remove first match
<code>pop(i)</code>	Remove by index
<code>clear()</code>	Remove all items
<code>reverse()</code>	Reverse in place

- Most methods mutate the list in place

Sorting: sort vs sorted

- `sort()` modifies the list in place

```
nums = [3, 1, 2]
nums.sort()                # nums becomes [1, 2, 3]
new_nums = sorted(nums, reverse=True)
```

Sorting: sort vs sorted

- `sort()` modifies the list in place
- `sorted(list)` returns a new list

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nums = [3, 1, 2]
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```


Sorting: sort vs sorted

- `sort()` modifies the list in place
- `sorted(list)` returns a new list
- Use `sorted` when you must preserve the original

```
nums = [3, 1, 2]
nums.sort()                                # nums becomes [1, 2, 3]
new_nums = sorted(nums, reverse=True)
```

List Operations

- Concatenation: $[1,2] + [3,4]$

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- Concatenation: `[1,2] + [3,4]`
- Repetition: `[0] * 3`
- Membership: `5 in nums`
- Length: `len(nums)`

Pitfall: Aliasing vs Copy

```
a = [1, 2, 3]
b = a          # alias (same object)
b.append(99)
print(a)       # [1, 2, 3, 99]
```

■ Fix: create an independent list copy

```
b = a.copy()   # or: a[:]
```

List Comprehension

- Compact way to build lists

```
squares = [x*x for x in range(6)]
```

List Comprehension

- Compact way to build lists
- Pattern: `[expr for item in iterable if cond]`

```
squares = [x*x for x in range(6)]
```


Comprehension Example

```
nums = [3, 4, 7, 8, 10]
result = [n*n for n in nums if n % 2 == 0]
print(result)  # [16, 64, 100]
```

- Combine filtering and transformation in one line

Comprehensions: Filtering vs if-else

- Filter (keep/remove elements): `[x for x in data if cond]`

```
nums = [1, 2, 3, 4, 5, 6]
evens = [n for n in nums if n % 2 == 0]
labels = ["even" if n % 2 == 0 else "odd" for n in nums]
```

Comprehensions: Filtering vs if-else

- Filter (keep/remove elements): `[x for x in data if cond]`
- Choose a value (same length): `[A if cond else B for x in data]`

```
nums = [1, 2, 3, 4, 5, 6]
evens = [n for n in nums if n % 2 == 0]
labels = ["even" if n % 2 == 0 else "odd" for n in nums]
```

Nested Comprehensions (Flattening)

- Useful for 2D data (lists of lists)

```
matrix = [[1, 2, 3], [4, 5, 6]]  
flat = [x for row in matrix for x in row]  
print(flat)    # [1, 2, 3, 4, 5, 6]
```

Nested Comprehensions (Flattening)

- Useful for 2D data (lists of lists)
- Reads like two nested loops, left to right

```
matrix = [[1, 2, 3], [4, 5, 6]]  
flat = [x for row in matrix for x in row]  
print(flat)    # [1, 2, 3, 4, 5, 6]
```

Worked Example: Marks Analyzer (Core Steps)

```
marks = [72, 88, 95, 64, 51, 43, 77]

top_three = sorted(marks, reverse=True)[:3]
pass_mark = 50
passing = [m for m in marks if m >= pass_mark]
```

- Use sorted to avoid mutating the input list

Worked Example: Marks Analyzer (Core Steps)

```
marks = [72, 88, 95, 64, 51, 43, 77]

top_three = sorted(marks, reverse=True)[:3]
pass_mark = 50
passing = [m for m in marks if m >= pass_mark]
```

- Use sorted to avoid mutating the input list
- Use a comprehension to filter by a rule

Nested Lists

- Lists can contain other lists (2D structures)

```
matrix = [[1, 2, 3], [4, 5, 6]]  
print(matrix[1][2])    # 6
```


Nested Lists

- Lists can contain other lists (2D structures)
- Access with two indices: `matrix[row][col]`

```
matrix = [[1, 2, 3], [4, 5, 6]]  
print(matrix[1][2])    # 6
```

Pitfall: Modify While Iterating

- Removing items while iterating can skip elements

```
nums = [1, 2, 3, 4, 5]
nums = [n for n in nums if n % 2 == 1]
print(nums)  # [1, 3, 5]
```

Pitfall: Modify While Iterating

- Removing items while iterating can skip elements
- Prefer building a new list (filter) or iterate over a copy

```
nums = [1, 2, 3, 4, 5]
nums = [n for n in nums if n % 2 == 1]
print(nums)    # [1, 3, 5]
```

Demo: Student Marks Analyzer

- Load a list of marks

Script: `demo/list_marks_analyzer.py`

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- Compute average and top 3 scores

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Demo: Student Marks Analyzer

- Load a list of marks
- Compute average and top 3 scores
- Filter passing marks using a comprehension
- Extension: compute median as a challenge

Script: `demo/list_marks_analyzer.py`

Checkpoint 1

Which of these returns a new list instead of mutating?

1 append

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2 sort

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Which of these returns a new list instead of mutating?

- 1 append
- 2 sort
- 3 sorted

Checkpoint 2

Write a list comprehension to square only even numbers from `nums`
`= [1,2,3,4,5,6]`.

Checkpoint 3

What is the difference between these two comprehensions?

1 `[x for x in data if cond]`

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What is the difference between these two comprehensions?

- 1 `[x for x in data if cond]`
- 2 `[A if cond else B for x in data]`

Think-Pair-Share

When is a **for-loop** clearer than a list comprehension? Give a short example and justify your choice.

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- Lists are ordered, mutable collections
- Indexing/slicing enable flexible access
- Methods like `append` and `sort` mutate lists
- Comprehensions build lists concisely

Exit Question

What slice returns the last three items of a list data?